



Crack the Hash

Offensive Security Write-up Autor: Natalia Farias Data: 17 de Abril de 2026

Sinopse

O *crack the hash* é um desafio que explora a quebra de códigos hash.

1. Nível 1

Primeiro Código: 48bb6e862e54f2a795ffc4e541caed4d

O primeiro passo foi identificar o tipo de hash com hash-identifier:

```
kali@kali:~$ hashid 48bb6e862e54f2a795ffc4e541caed4d
Analyzing '48bb6e862e54f2a795ffc4e541caed4d'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
```

Ao descobrir que possivelmente poderia ser MD5, foi utilizado hashcat com a tag “m” do MD5 que é igual a 0, ou seja, -m 0.

```
kali@kali:~/Downloads$ hashcat -a 0 -m 0 48bb6e862e54f2a795ffc4e541caed4d /usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELoc, SPIR-V, LLVM 18.1.8, SLEEP, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]

* Device #01: cpu-haswell-12th Gen Intel(R) Core(TM) i5-1235U, 1592/3184 MB (512 MB allocatable), 2MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Early-Skip
* Not-Salted
* Not-Iterated

Stopped. Sat Apr 11 13:17:27 2026
easy
```

O código retornado foi: “easy”



Segundo código: CBFDAC6008F9CAB4083784CBD1874F76618D2A97

O primeiro passo foi identificar o tipo de hash com hash-identifier:

```
(kali㉿kali)-[~]
$ hashid CBFDAC6008F9CAB4083784CBD1874F76618D2A97
Analyzing 'CBFDAC6008F9CAB4083784CBD1874F76618D2A97'
[+] SHA-1
[+] Double SHA-1
[+] RIPEMD-160
[+] Haval-160
[+] Tiger-160
[+] HAS-160
[+] LinkedIn
[+] Skein-256(160)
[+] Skein-512(160)
```

Depois foi utilizado hashcat com a tag -m 100, tag do SHA-1

```
(kali㉿kali)-[~]
$ hashcat -m 100 -a 0 'CBFDAC6008F9CAB4083784CBD1874F76618D2A97' --show
cbfdac6008f9cab4083784cbd1874f76618d2a97:password123
```

Resultado: password123

Terceiro Código: 1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032

O primeiro passo foi identificar o tipo de hash com hash-identifier que retornou SHA2-256.

Utilizando a tag -m 1400 foi encontrado o resultado: letmein

```
Session Actions Edit View Help
$ hashcat -m 1400 -a 0 1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032 /usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEP, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]

* Device #01: cpu-haswell-12th Gen Intel(R) Core(TM) i5-1235U, 1829/3659 MB (512 MB allocatable), 4MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Early-Skip
* Not-Salted
* Not-Iterated
* Single-Hash
* Single-Salt
* Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90C

Most memory allocated for this attack: 513 MB (1171 MB free)

Dictionary cache hit:
* Filename ..: /usr/share/wordlists/rockyou.txt
* Passwords ..: 14344385
* Bytes ..: 139921507
* Keyspace ..: 14344385

1c8bfe8f801d79745c4631d09fff36c82aa37fc4cce4fc946683d7b336b63032:letmein
```

Quarto Código: \$2y\$12\$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom



Ao utilizar o hash id para este código foi retornado um erro. Para identificar o tipo de hash foi consultado a lista de tipos de hash do hashcat para identificar através das iniciais.

Foi utilizado hashcat com -m 3200, bcrypt \$2*\$, Blowfish. E uma filtragem da wordlist rockyou com apenas palavras de 4 letras minúsculas, seguindo a dica do TryHackMe.

Resultado: Bleh

```

kali@kali: ~/Downloads
Session Actions Edit View Help
Candidate.Engine.: Host Generator + PCIE
Candidates.#01...: leon → Jeep
Hardware.Mon.#01.: Util: 78%

[s]tatus [p]ause [b]ypass [c]heckpoint [f]inish [q]uit ⇒ s
Session.....: hashcat
Status.....: Running
Hash.Mode.....: 3200 (bcrypt $2*$, Blowfish (Unix))
Hash.Target.....: $2y$12$Dwt1BZj6pcyc3Dy1FW25ieeUznr71EeNkJkUlypTsgbX...8wsRom
Time.Started.....: Sat Apr 11 14:16:30 2026 (40 secs)
Time.Estimated...: Sat Apr 11 14:34:04 2026 (16 mins, 54 secs)
Kernel.Feature...: Pure Kernel (password length 0-72 bytes)
Guess.Base.....: File (list.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 8 H/s (12.87ms) @ Accel:4 Loops:32 Thr:1 Vec:1
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 320/8541 (3.75%)
Rejected.....: 0/320 (0.00%)
Restore.Point....: 320/8541 (3.75%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:1312-1344
Candidate.Engine.: Host Generator + PCIE
Candidates.#01...: carl → gurl
Hardware.Mon.#01.: Util: 79%

$2y$12$Dwt1BZj6pcyc3Dy1FW25ieeUznr71EeNkJkUlypTsgbX1H68wsRom:bleh

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 3200 (bcrypt $2*$, Blowfish (Unix))
Hash.Target.....: $2y$12$Dwt1BZj6pcyc3Dy1FW25ieeUznr71EeNkJkUlypTsgbX...8wsRom
Time.Started.....: Sat Apr 11 14:16:30 2026 (59 secs)
Time.Estimated...: Sat Apr 11 14:17:29 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-72 bytes)
Guess.Base.....: File (list.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 8 H/s (11.03ms) @ Accel:4 Loops:32 Thr:1 Vec:1

```

Quinto código: 279412f945939ba78ce0758d3fd83daa

Utilizado hashid para identificar o tipo de hash.

```

kali@kali: ~/Downloads
* Device #01: cpu-haswell-12th Gen Intel(R) Core(TM) i5-1235U, 1829/3659 MB (512 MB allocatable), 4MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 72
Minimum salt length supported by kernel: 0
Maximum salt length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.
For more information, see https://hashcat.net/faq/potfile

Started: Sat Apr 11 14:17:48 2026
Stopped: Sat Apr 11 14:17:48 2026

(kali@kali)~[~/Downloads]
$ hashid "279412f945939ba78ce0758d3fd83daa"
Analyzing '279412f945939ba78ce0758d3fd83daa'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus-Notes/Domino 5
[+] Skype
[+] Snefru-128
[+] NTLM
[+] Domain Cached Credentials
[+] Domain Cached Credentials 2
[+] DNSSEC(NSEC3)
[+] RAdmin v2.x

```

Search for a tool

e.g. type "vudokur"

BROWSE THE FULL CODE TOOLS LIST

Results

Eternity22

MD4

Informatics , Algorithm , Hashing Function , MD4

MD4 DECODER

- MD4 HASH [279412f945939ba78ce0758d3fd83daa]
- OPTIONS
- SALT PREFIXED MD4(SALT+WORD)
- SALT SUFFIXED MD4(WORD+SALT)

Summary

- MD4 Decoder
- MD4 Encoder
- What is MD4? (Definition)
- How to encrypt in MD4?
- How to decrypt MD4 cipher?
- How to recognise MD4 ciphertext?
- What is a MD4 collision?
- What does the MD4 acronym mean?
- When was MD4 invented?

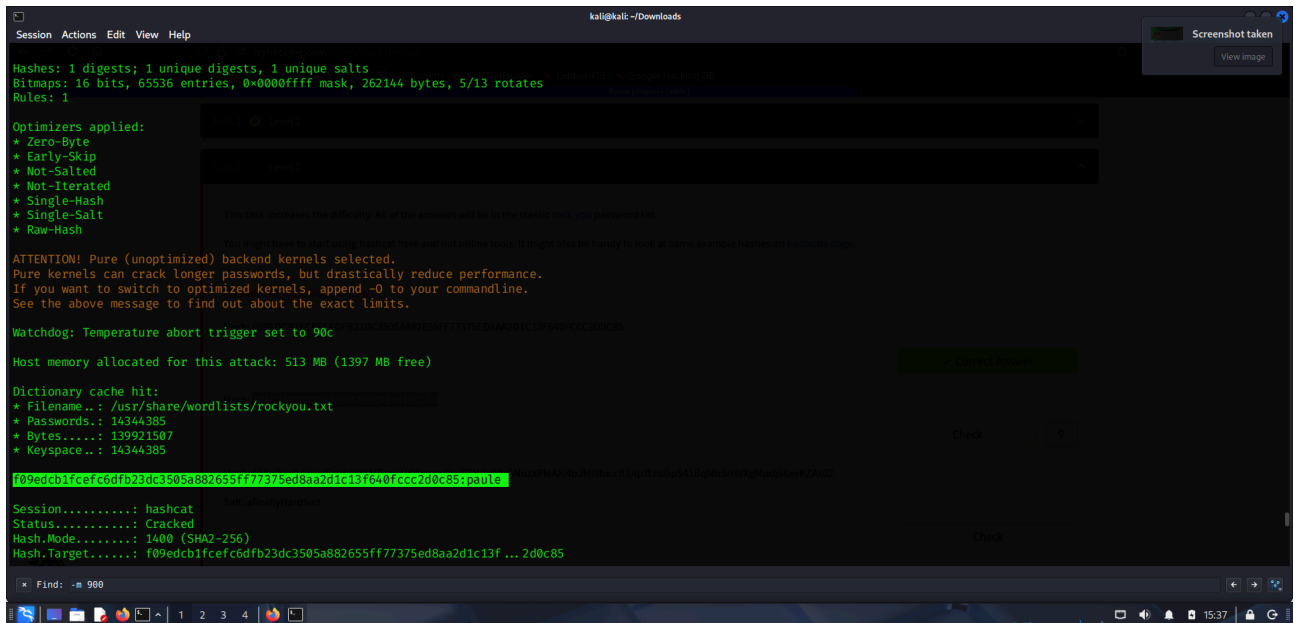
Similar pages

Feedback

Primeiro foi utilizado hash-identifier para descobrir o tipo de código, retornando SHA-256 nas possibilidades.

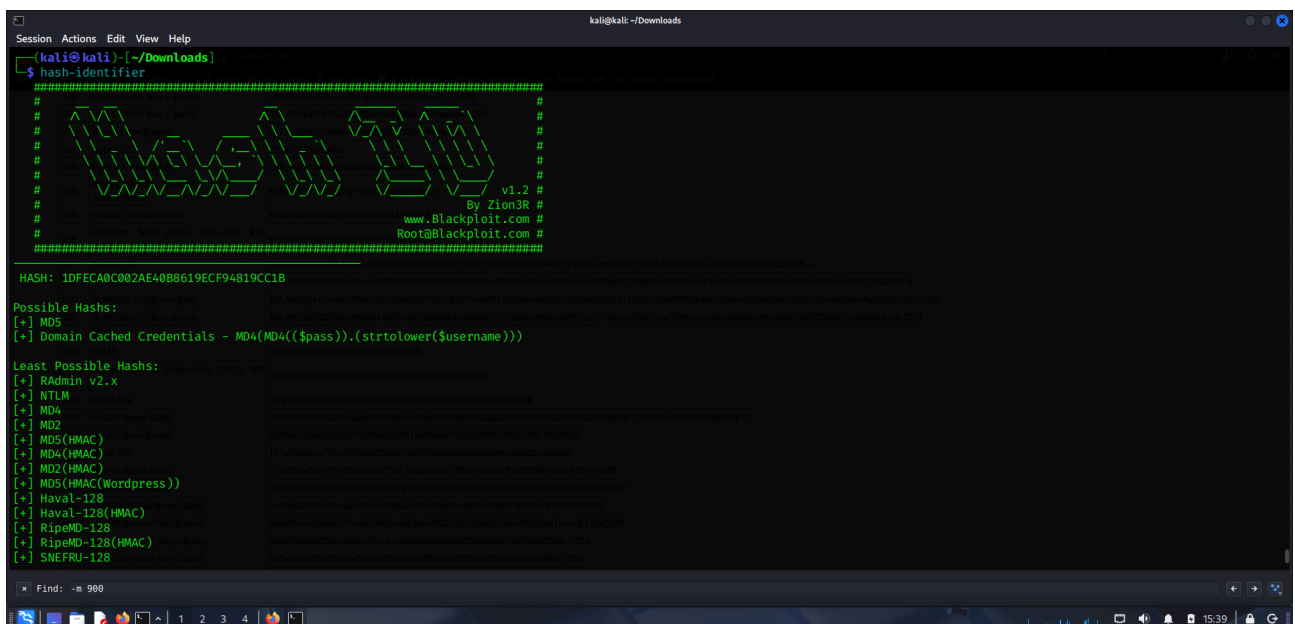
[illegible]

4



Segundo código: 1DFECA0C002AE40B8619ECF94819CC1B

Ao utilizar hashid foi encontrado: n63umy8lkf4i.



Com hashcat foi encontrado o resultado:



```

kali@kali:~$ hashcat -m 0 -a 0 1DFECA0C002AE40B8619ECF94819CC1B /usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]

* Device #01: cpu-haswell-12th Gen Intel(R) Core(TM) i5-1235U, 1829/3659 MB (512 MB allocatable), 4MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.
For more information, see https://hashcat.net/faq/potfile

Started: Thu Apr 16 19:47:29 2026
Stopped: Thu Apr 16 19:47:30 2026

kali@kali:~$ hashcat -m 0 -a 0 1DFECA0C002AE40B8619ECF94819CC1B /usr/share/wordlists/rockyou.txt --show
1dfeca0c002ae40b8619ecf94819cc1b:n63umy8lkf41

kali@kali:~$

```

Terceiro Código:

\$6\$aReallyHardSalt\$6WKUTqzq.UQQmrmOp/T7MPpMbGNnzXPMAXi4bJMI9be.cfi3/qxlf.hsGpS41BqMhSrHVXgMpdjS6xeKZAs02

Foi identificado o tipo de hash na lista de tipo de hashcat.

1750	HMAC-SHA512 (key = \$pass)	94cb9e31137913665d5ba7b058e10be5f050cc356062a2c9679ed0ad6119648e7be620e9d4e1199220cd02b9eb2b1c78234fa1000c728f82b9f14ed82c1976:1234
1760	HMAC-SHA512 (key = \$salt)	7cce966f5503e292a513811238d071971ad5442488f340f98e379b3aaea2f3778e3e732cc2f7bdc04f34460eebf6f8cb77da32d2f25500c09160dd3b7d2af6b:1234
1770	sha512(utf16le(\$pass))	79bba09eb9354412d0f2c037c22a777b8bf549ab12449677d5b25faa839e4378d8f6a11aceb6d9413977ae5ad5d011568baad2de4f998d7594ace916eda83697
1800	sha512crypt(\$\$, SHA512 (libcrypt))	\$6\$qtMgClyO2dOWB37F\$jhvCX1SdsCA0OZmoRVAPnWSwuPmhVhXIMJIKlaaczFlwWLDZDVf8U3WcHaashcatVp2WShcEIL8QZCbt
2000	STDOUT	n/a
2100	Domain Cached Credentials 2 (DCC2), MS Cache 2	\$DCC2\$10240#tomfe4e938d12fe5974dc42a90120bd9c90f
2400	Cisco-PIX MD5	dPRVnolJmJlJHXYOT8nk

Usando hashcat com o salto informado pelo tryhackme foi encontrado: waka99

The screenshot shows a Kali Linux terminal window with the following content:

```
kali@kali: ~/Downloads
```

Session Actions Edit View Help

kali@kali: ~/Downloads

```
Guess.Base.....: File (list6.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 727 H/s (3.12ms) @ Accel:33 Loops:124 Thr:1 Vec:4
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 9372/1949232 (0.48%)
Rejected.....: 0/9372 (0.00%)
Restore.Point...: 9372/1949232 (0.48%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:2480-2604
Candidate.Engine.: Device Generator
Candidates.#01...: punjab → 050306
Hardware.Mon.#01.: Util: 71%
```

[s]tatus [p]ause [b]ypass [c]heckpoint [f]inish [q]uit ⇒ s

```
Session.....: hashcat
Status.....: Running
Hash.Mode.....: 1800 (sha512crypt $6$, SHA512 (Unix))
Hash.Target....: $6$ReallyHardSalt$GWKUTzqz.UQqmm0p/T7MPpMbGnNzXPM...ZAs02.
Time.Started...: Sat Apr 11 15:47:36 2026 (3 mins, 21 secs)
Time.Estimated...: Sat Apr 11 16:28:04 2026 (37 mins, 7 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (list6.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 802 H/s (3.02ms) @ Accel:33 Loops:124 Thr:1 Vec:4
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 161568/1949232 (8.29%)
Rejected.....: 0/161568 (0.00%)
Restore.Point...: 161568/1949232 (8.29%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:3348-3472
Candidate.Engine.: Device Generator
Candidates.#01...: ciurea → chip69
Hardware.Mon.#01.: Util: 75%
```

\$6\$ReallyHardSalt\$GWKUTzqz.UQqmm0p/T7MPpMbGnNzXPMAX14bJM19be.cf.i3/qxIf.hs6pS418qMhSrHVXgMpdjS6xeKZAs02.:waka99

Quarto Código: e5d8870e5bdd26602cab8dbe07a942c8669e56d6

Hash-id:

[illegible]

Utilizando a dica do tryhack me foi utilizado a tag -m 160 do HMAC-SHA1. Resultado:
481616481616



```
kali@kali: ~/Downloads
Session Actions Edit View Help
kali@kali: ~/Downloads
Candidate.Engine.: Device Generator
Candidates.#01...: kristenanne → $HEX[042a0337c2a156616d6f732103]
Hardware.Mon.#01.: Util: 39%

Started: Sat Apr 11 18:33:23 2026
Stopped: Sat Apr 11 18:33:45 2026

(kali@kali) [~/Downloads]
$ hashcat -m 160 -a 0 'e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme' /usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPTR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
```

```
kali@kali: ~/Downloads
Session Actions Edit View Help
kali@kali: ~/Downloads
* Bytes.....: 139921507
* Keyspace...: 14344385

[s]tatus [p]ause [b]ypass [c]heckpoint [f]inish [q]uit ⇒ s

Session.....: hashcat
Status.....: Running
Hash.Mode.....: 160 (HMAC-SHA1 (key = $salt))
Hash.Target.....: e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme
Time.Started.....: Sat Apr 11 18:34:34 2026 (2 secs)
Time.Estimated...: Sat Apr 11 18:34:41 2026 (5 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 1972.9 kH/s (0.54ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 4419584/14344385 (30.81%)
Rejected.....: 0/4419584 (0.00%)
Restore.Point....: 4419584/14344385 (30.81%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: rachand → raain
Hardware.Mon.#01.: Util: 37%

e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme:481616481616

You started a streak, let's go!
How long can you keep your skills sharp?
```